MASTER IDEATION & SYSTEM SPECIFICATION DOCUMENT (IDS V1.0)

SSM Events & Ticketing Platform

Project Codename: SSM Events & Ticketing (Sasaram & Bihar Hyper-Local Platform)

Target Market: Sasaram (Rohtas), Dehri-on-Sone, Gaya, Patna & Tier-2/3 Bihar Cities

Business Model: Hyper-Local Event Listing + Box-Office QR Ticketing (7% Flat Commission)

Target Infrastructure Cost: ₹0 / Month (100% Free-Tier Cloud Architecture)

1. EXECUTIVE SUMMARY & MARKET OPPORTUNITY

MARKET THESIS

Major ticketing platforms (BookMyShow, Paytm Insider, Zomato District) ignore Tier-2 and Tier-3 cities in Bihar or charge organizers 10% to 15% commissions with slow corporate onboarding and 7-day post-event payouts. **The Solution:** A fast, mobile-first, no-login UPI ticketing platform built specifically for local events in Sasaram and Bihar—covering coaching institute seminars and scholarship tests, New Year & Dandiya parties, college fests, school annual functions, local concerts, and community workshops.

- **Lower Commission (7% Total vs. 10–15% on Traditional Platforms):** Organizers keep 93% of ticket value—or pay **0% commission** if passed to the buyer as a convenience fee.
- **Zero Tech Friction:** Organizers do not need to configure API keys; they simply request withdrawals to their Bank Account or UPI ID directly from their organizer dashboard.
- **Fast T+2 Settlements & Hybrid Sales:** Rapid T+2 bank settlements paired with built-in Offline Cash-at-Counter Promoter Passes for physical reception desks.

2. UNIT ECONOMICS & FINANCIAL MODEL

4.64% NET MARGIN

2.1 Per-Ticket Commission Breakdown (Base Case: ₹800.00 Ticket)

Ticket Base Price: ₹800.00 | **Platform Commission (7%):** ₹56.00 | **Razorpay Standard Gateway Fee (2%):** ₹16.00 | **GST on Razorpay Fee (18% of ₹16):** ₹2.88 (automatically deducted by Razorpay; effective gateway fee is 2.36%) | **Your Net Profit per Ticket (4.64%): ₹37.12**.

Net Profit per Ticket = Gross Fee (7.00%) − Gateway Cut incl. GST (2.36%) = 4.64% of Ticket Price

2.2 Revenue Projections by Event Scale

Event Type & Scale	Ticket Price	Attendance	Gross Ticket Volume (GMV)	Total 7% Fee Collected	Razorpay Cut (2.3
Coaching Seminar	₹100	500 Students	₹50,000	₹3,500	
College Fest / Show	₹300	800 Attendees	₹2,40,000	₹16,800	
Medium New Year Party	₹800	500 Attendees	₹4,00,000	₹28,000	
Large New Year / Dandiya	₹800	1,000 Attendees	₹8,00,000	₹56,000	

3. OPEN-SOURCE REPOSITORIES & REVERSE-ENGINEERING MATRIX

ARCHITECTURE FOUNDATION

Rule: Do not merge different programming languages together. Use **Next.js + TypeScript + Supabase** as your single unified foundation, and reverse-engineer specific UI patterns and workflows from these six verified open-source repositories:

Project & GitHub URL	Tech Stack	License	Exact Features to Reverse-Engineer / Copy
1. Sale-Ticket (TicketFlow) `github.com/abdelhamid11111111/sale-ticket`	Next.js, TypeScript, Prisma, Supabase, Tailwind	MIT	**Your Base Starter Template:** Copy the 2-step no-login checkout UI (Buyer Info + Quantity → Pay), session-based "My Tickets" page, and `/admin` Revenue & Orders dashboard.
2. Tickity `github.com/akash202004/tickity`	Next.js 14, Tailwind, QR Tickets	Open Source	**Queue & Ticket Card UI:** Copy the visual QR ticket card layout and real-time waiting-room queue logic for high-demand ticket drops.
3. Hi.Events `github.com/HiEventsDev/Hi.Events`	PHP (Laravel), React, PostgreSQL	AGPL-3.0	**Gate Check-In Scanner & Promo Codes:** Study how their browser-based mobile camera QR scanner works (`checkin-web`) and how tiered ticket inventories are structured.
4. Revel `github.com/letsrevel/revel-backend` `github.com/letsrevel/revel-frontend`	Python (Django) + SvelteKit	MIT	**Coaching & Community Features:** Reverse-engineer their custom RSVP questionnaire forms (e.g., asking students for Batch / Class / Roll No during checkout).
5. Gathio `github.com/lowercasename/gathio`	Node.js, Express, MongoDB	AGPL-3.0	**60-Second Event Creation Flow:** Copy the ultra-minimalist single-page event creator and WhatsApp-friendly OpenGraph preview cards.
6. Pollaris `pollaris.org`	Web Polling Tool	Open Source	**Live Audience Q&A / Voting Upsell:** Add a simple live poll tab on event pages so coaching institutes and college fests can run live audience voting during the event.

4. THE ₹0/MONTH CLOUD ARCHITECTURE & TRAP MITIGATION

100% FREE TIER

4.1 Infrastructure Stack (100% Free-Tier Cloud Components)

- **Frontend & Serverless Backend API — Vercel (Free Hobby Tier):** Runs Next.js App Router with zero-cold-start serverless routes.
- **Primary Database — Supabase (Free Tier):** 500MB PostgreSQL database capable of storing ~5,000+ events and 500,000+ ticket rows.
- **Concurrency & Gate Scanner Cache — Upstash Redis (Free Tier):** Prevents overselling when 50 people try to buy the last 5 tickets simultaneously; validates QR scans at the entry gate in under 50ms.
- **Poster & Ticket Image Hosting — Cloudinary (Free Tier):** Auto-compresses event posters to WebP format so pages load fast on slow 4G connections across Bihar.
- **DNS, CDN & DDoS Protection — Cloudflare (Free Tier):** Protects against traffic spikes and executes scheduled Cron Triggers at zero cost.
- **Payment Gateway — Razorpay (Primary) / Cashfree Payments (Backup):** Standard UPI & card processing, with Cashfree as backup for Easy Split.

4.2 Critical Free-Tier Traps & How We Solved Them

Trap 1: Why We Removed Render Free Tier

Trap 2: Preventing the Supabase 7-Day Pause

5. TAX, GST, LEGAL COMPLIANCE & ANTI-FREEZE RULES

REGULATORY & KYC

5.1 Why You Must NEVER Rotate Family Members' PAN / Bank Accounts

- **Income Tax AIS Tracking:** All bank accounts are linked to a PAN card's Annual Information Statement (AIS). Splitting money across 4 bank accounts under the same PAN still sums together automatically.
- **Razorpay Domain-Mismatch Freeze:** Switching Razorpay API keys to a family member's PAN on the same domain (`yourwebsite.com`) triggers Razorpay's aggregator fraud detection and freezes payouts for up to 180 days, while exposing family members to Income Tax scrutiny notices.

5.2 The Safe 2-Stage Legal Roadmap

Stage 1: Direct Collection (₹0 to ₹18L GMV) — 0–18 Lakhs

- **GST Threshold:** In Bihar, GST registration is only mandatory once annual turnover crosses ₹20 Lakhs.
- **Zero-GST Razorpay KYC Stack:** Register a free Udyam MSME Certificate (`udyamregistration.gov.in`, ₹0 cost, 10 mins with Aadhaar + PAN) and submit Razorpay's GST Non-Enrollment Declaration.
- **Collection Agent Agreement:** Sign a 1-page agreement with each organizer stating you act strictly as a ticketing collection agent where 93% of proceeds belong to the organizer and only the 7% fee is your gross revenue.

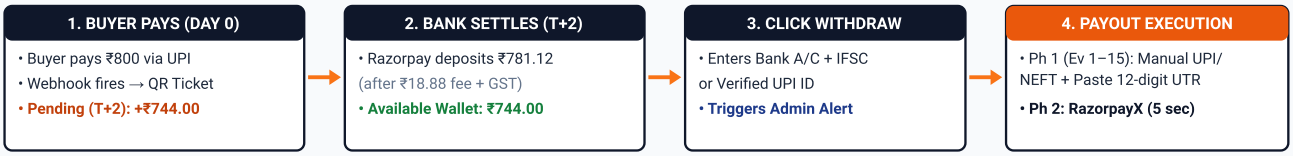
Stage 2: Automated Split (> ₹18L GMV) — Post-18 Lakhs

- **GST + Split Enablement:** Once you have processed ₹18 Lakhs in tickets (and earned ~₹83,500+ in net profit), spend ₹1,500 from your profits to register for GST and enable **Razorpay Route** or **Cashfree Easy Split**.
- **Gateway-Level Split:** From that point onward, the organizer's 93% share splits at the payment gateway level directly to the organizer's PAN and bank account, so only your 7% commission counts toward your turnover.

6. ORGANIZER WALLET, WITHDRAWAL & ANTI-SCAM WORKFLOW

T+2 PAYOUT ENGINE

6.1 How Organizers Withdraw Money (Zero Friction Flow — Base ₹800 Ticket)



6.2 Three Mandatory Financial Safety Rules

1. **Never Pay Out Unsettled Funds:** Only allow withdrawals on money that has already completed Razorpay's T+2 bank settlement.
2. **10%–20% Event-Completion Holdback:** Allow verified local organizers to withdraw 80% to 90% prior to the event (to pay DJ/venue/tent vendors), and hold back 10% to 20% until the morning after the event to protect buyers against last-minute cancellations.
3. **Auto-Generate Settlement Receipts:** Every payout generates a 1-page PDF statement showing Gross Ticket Collections, Less 7% Platform Fee, Net Payout Transferred, and the 12-digit Bank UTR Number for clean accounting.

7. BIHAR / TIER-2 CITY HYPER-LOCAL FEATURES (UNFAIR ADVANTAGE)

LOCAL MOAT

8. COMPLETE DATABASE SCHEMA & LAUNCH BLUEPRINT POSTGRESQL + ROADMAP

8.1 Supabase PostgreSQL Schema (`schema.sql`)

Complete production SQL DDL reconstructing the three core relational tables (Events & Organizer Wallet, Tickets & Orders, and Organizer Withdrawal Requests) alongside inline column specifications:

```
-- 1. Events & Organizer Wallet Table CREATE TABLE events ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), slug TEXT UNIQUE NOT NULL, -- e.g., 'sasaram-new-year-2027' title TEXT NOT NULL, venue TEXT NOT NULL, event_date TIMESTAMPTZ NOT NULL, ticket_price INTEGER NOT NULL, -- Base price in INR (e.g., 800) pass_fee_to_buyer BOOLEAN DEFAULT FALSE, -- FALSE: Absorb 7% | TRUE: Buyer pays ₹856 total_capacity INTEGER NOT NULL, tickets_sold INTEGER DEFAULT 0, organizer_name TEXT NOT NULL, organizer_whatsapp TEXT NOT NULL, wallet_pending_settlement NUMERIC DEFAULT 0, -- 93% share awaiting T+2 settlement wallet_available NUMERIC DEFAULT 0, -- Settled funds ready to withdraw wallet_total_withdrawn NUMERIC DEFAULT 0, -- Total already paid out to organizer scanner_pin TEXT NOT NULL, -- 4-digit PIN for gate security guard created_at TIMESTAMPTZ DEFAULT now() ); -- 2. Tickets & Orders Table CREATE TABLE tickets ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), event_id UUID REFERENCES events(id) ON DELETE CASCADE, buyer_name TEXT NOT NULL, buyer_whatsapp TEXT NOT NULL, payment_mode TEXT DEFAULT 'ONLINE', -- 'ONLINE' or 'CASH_PROMOTER' amount_paid NUMERIC NOT NULL, -- e.g., 800 or 856 organizer_share NUMERIC NOT NULL, -- e.g., 744 or 800 platform_fee NUMERIC NOT NULL, -- e.g., 56 (7%) razorpay_order_id TEXT UNIQUE, payment_status TEXT DEFAULT 'PENDING', -- 'PENDING', 'PAID', 'REFUNDED' qr_token TEXT UNIQUE, -- Cryptographic token encoded in QR (SSM-XXXX) is_checked_in BOOLEAN DEFAULT FALSE, -- Flips to TRUE at entry gate created_at TIMESTAMPTZ DEFAULT now() ); -- 3. Organizer Withdrawal Requests Table CREATE TABLE withdrawal_requests ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), event_id UUID REFERENCES events(id) ON DELETE CASCADE, amount NUMERIC NOT NULL, payout_destination TEXT NOT NULL, -- UPI ID or Bank Account + IFSC status TEXT DEFAULT 'REQUESTED', -- 'REQUESTED', 'COMPLETED', 'REJECTED' bank_utr_number TEXT, -- 12-digit bank reference number created_at TIMESTAMPTZ DEFAULT now() );
```

8.2 Step-by-Step 14-Day Launch Roadmap

Sprint Window	Milestone Focus	Concrete Engineering & Go-To-Market Deliverables
Days 1–3	**Base Setup**	Clone `abdelhamid11111111/sale-ticket`, connect to your free Supabase PostgreSQL database, and execute the `schema.sql` blueprint above.
Days 4–6	**Razorpay Test Mode**	Create a Razorpay account, switch to Test Mode, and wire up `/api/create-order` and `/api/webhook/razorpay`. Test simulated UPI payments using `success@razorpay`.
Days 7–9	**QR Ticket & Gate Scanner**	Add HMAC-signed `qrcode` generation on the ticket confirmation screen and build the password-protected `/scan/[event-id]` mobile camera scanner page using `html5-qrcode`.
Days 10–11	**Organizer Wallet & Cash Promoter Mode**	Build the Organizer Dashboard displaying Tickets Sold, Pending/Available Balance, the "Request Withdrawal" button, and the offline "Issue Cash Ticket" button.
Days 12–14	**KYC & First Local Pitch in Sasaram**	Register your free Udyam MSME Certificate, complete Razorpay Live KYC with the GST Non-Enrollment Declaration, and pitch your first local coaching seminar or party organizer in Sasaram with a live mobile demo.